

Khulna University of Engineering & Technology  
B.Sc. Engineering 2<sup>nd</sup> Year 1<sup>st</sup> Term (Regular) Examination, 2019  
Department of Materials Science and Engineering

**MSE 2101**  
**Crystallography**

Time: 3 hours

Full Marks: 210

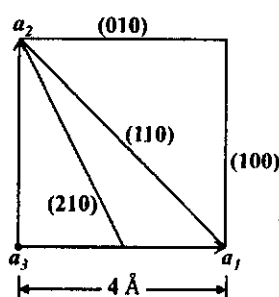
- N.B. (i) Answer **ANY THREE** questions from each section in separate scripts.  
(ii) Figure in the right margin indicates full marks  
(iii) Use Wulff net and Tracing paper where needed. Both are available on examination hall upon request.

**Section A**

(Answer **ANY THREE** questions from this section in **answer script A**)

- Q 1. (a) In 2D, the reciprocal lattice of an oblique crystal is also oblique. Explain with necessary diagrams. (08)

- (b) Draw the reciprocal lattice of the following real lattice. Include all the necessary dimensions and direction vectors. (10)

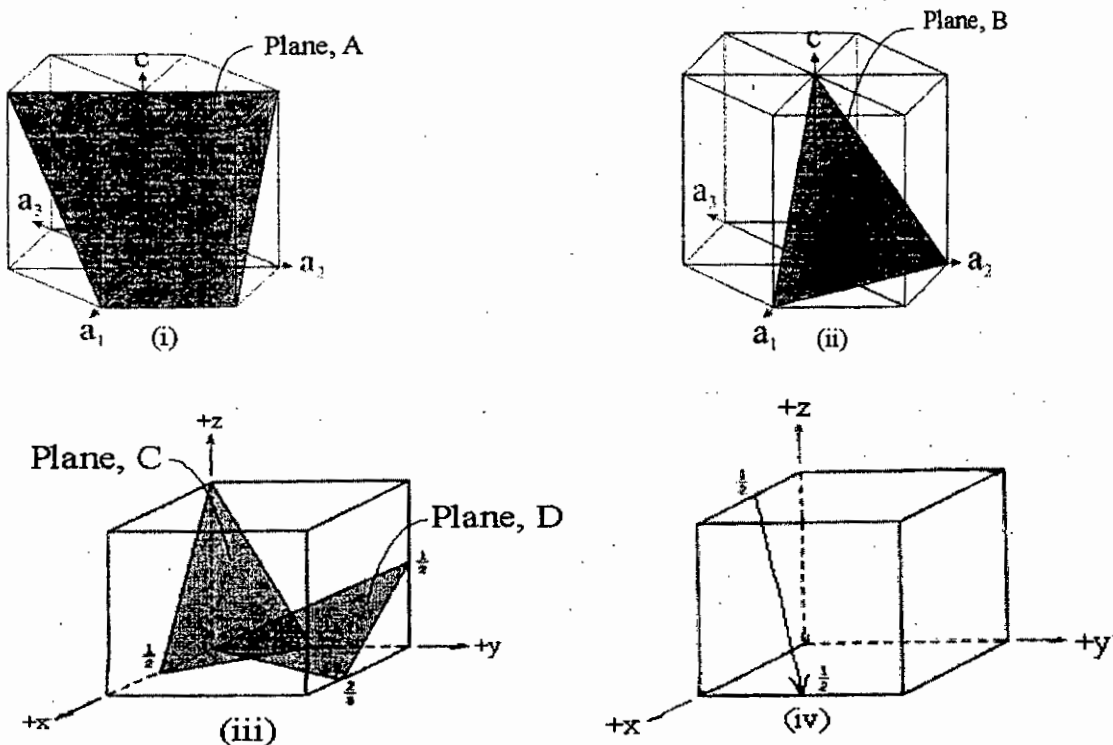


- (c) Write down the basis vectors of a cubic I-lattice and mathematically prove that the reciprocal lattice of an I-lattice is actually F-lattice. (10)
- (d) Write short note on any **ONE** of the following: (i) Wigner-Seitz Cell (ii) Brillouin Zone (07)
- Q 2. (a) We do not get diffracted x-ray beam for both (001) and (002) planes in BCC crystal. Explain with necessary equations. (07)
- (b) How many orders of reflection are possible if the interplanar spacing is 1.506 Å and the wavelength is 0.724 Å? At what angle will they occur? (10)
- (c) A metallic sample gives intense peaks in the diffraction pattern for the following ( $2\theta$ ) angles: 44.5859°, 64.9805°, 82.2971°, 99.1627°, 116.5862°, 137.3325° and 180°. Find out the lattice parameter of the sample and comment on the crystal structure. (18)
- Q 3. (a) What crystallographic information does the Scattering Intensity versus Scattering Angle curve of XRD provide? (06)
- (b) What are the advantages of neutron diffraction over x-ray diffraction? (08)
- (c) The crystallite size of powdered Bismuth ferrite sample at  $2\theta = 31.9291^\circ$  is 332 Å. Determine the FWHM in radian when no strain is introduced. (10)
- If the lattice strain is 0.42218%, then comment on the change in the shape of the peak.  
Here,  $B = \frac{0.94 \lambda}{D \cos \theta} + 4e \tan \theta$  (the symbols have their usual meaning in crystallography)
- (d) Write down the differences between electron diffraction and neutron diffraction. Which one will you prefer for the determination of the magnetic structure of a powdered sample? (11)
- Q 4. (a) Why is stereographic projection important in crystallography? (10)
- (b) (i) Draw the stereographic projection of a cubic lattice on (001) plane using the Wulff net. (07)
- (ii) Determine the intersection of the traces of plane (0 $\bar{1}$ 1) and (111) and indicated it as pole 'A'. Again, determine the intersection of the traces of plane (1 $\bar{1}$ 1) and (110) and indicated it as pole 'B' (10)
- (iii) Determine the Miller indices of pole A and B. Calculate the angular distance between pole A and pole B using the Wulff net. (08)

## Section B

(Answer **ANY THREE** questions from this section in **answer script B**)

- Q 5. (a) Describe the voids present in a FCC and BCC crystal. The FCC and BCC crystals have 26% and 32% of unoccupied space in their respective unit cell, but the maximum size of atom that can be hold into FCC interstices is larger than the BCC structure, explain the reason. (10)
- (b) Show that, the relationship of base-edge length ( $a$ ) of HCP unit cell and its height ( $c$ ) is,  $c=1.633a$ . Henceforth, calculate the APF of the HCP unit cell. (15)
- (c) Calculate the planner density of (111) plane of FCC unit cell. (10)
- Q 6. (a) What is motif? How does motif differ from a lattice point during formation of a crystal structure? (10)
- (b) Calculate the critical radius ratio of a foreign atom that can be fitted into a tetrahedral void of a FCC structure. (10)
- (c) Why is it not possible to have 5 or greater than 6-fold rotational symmetry for a 2-D crystal system? Justify your answer with example. (15)
- Q 7. (a) What is unit cell? "The choice of unit cell is not unique"-explain the statement with appropriate example. (10)
- (b) "Honeycomb lattice never forms the Bravais lattice"—justify the statement. (07)
- (c) Explain, why a larger interstitial atom preferentially accommodates in the tetrahedral site rather than the octahedral site of a BCC crystal. (08)
- (d) What are the differences between FCC and HCP crystal lattice though both of them are constructed from a similar close packing. (10)
- Q 8. (a) Explain, why there is no base cantered, C-lattice for tetragonal crystal system? (10)
- (b) Write the steps of determination of Miller-Indices for direction with example. (10)
- (c) Determine the Miller-Indices of the following planes/ directions: (3×5)



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**Math 2127**  
**Vector Analysis and Linear Algebra**

Time: 3 hours

Full Marks: 210

- N.B. (i) Answer **ANY THREE** questions from each section in separate scripts.  
(ii) Figure in the right margin indicates full marks

**Section A**

(Answer **ANY THREE** questions from this section in **answer script A**)

- Q 1. (a) Find the unit tangent vector to the point where  $t=2$  on the curve (08)  
$$x = t^2 + 1, y = 4t - 3, z = 2t^2 - 6t$$
- (b) A particle moves along the curve  $x = 2t^2, y = t^2 - 4t, z = 3t - 5$ , where  $t$  is the time. Find the components of its velocity and acceleration at  $t = 1$  in the direction  $\underline{i} - 3\underline{j} + 2\underline{k}$ . (12)
- (c) Find the directional derivative of  $\phi = x^2yz + 4xz^2$  at  $(1, -2, -1)$  in a direction toward the point  $(2, -1, -2)$ . (08)
- (d) Evaluate  $\nabla^2(\ln r)$ . (07)
- Q 2. (a) Find the equations of tangent plane and normal line to the surface  $z = x^2 + y^2$  at the point  $(2, -1, 5)$ . (12)
- (b) Test whether the vector field  $\underline{F} = (z^3 + y^2 \cos x)\underline{i} - (4 - 2y \sin x)\underline{j} + (2 + 3xz^2)\underline{k}$  is conservative or not. If it is conservative, find its scalar potential. (12)
- (c) Find the total work done in moving a particle in the force field  $\underline{F} = (2y + 3)\underline{i} + xz\underline{j} + (yz - x)\underline{k}$  along the path  $C$ , where  $C$  is the straight line joining  $(0, 0, 0)$  to  $(2, 1, 1)$ . (11)
- Q 3. (a) Evaluate  $\iint_S \underline{A} \cdot \underline{n} dS$  over the surface  $S$  of the region bounded by the cylinder  $x^2 + z^2 = 9$ ,  $x = 0, y = 0, z = 0$  and  $y = 8$ , if  $\underline{A} = 6z\underline{i} + (2x + y)\underline{j} - x\underline{k}$ . (15)
- (b) Verify the divergence theorem for  $\underline{A} = 4x\underline{i} - 2y^2\underline{j} + z^2\underline{k}$  taken over the region bounded by  $x^2 + y^2 = 4, z = 0$  and  $z = 3$ . (20)
- Q 4. (a) Evaluate  $\oint_C (xy + y^2)dx + x^2dy$ , where  $C$  is the closed region bounded by  $y = x$  and  $y = x^2$  by using Green's theorem. (11)
- (b) Define curvilinear coordinates of a point. Express the velocity  $\underline{v}$  and acceleration  $\underline{a}$  of a particle in cylindrical coordinates. (12)
- (c) Derive an expression for  $\nabla\phi$  in orthogonal curvilinear coordinates. (12)

### Section B

(Answer ANY THREE questions from this section in answer script B)

- Q 5. (a) Define with example: (i) Augmented matrix of a system, (ii) Elementary matrix, (iii) Orthogonal matrix. (12)

- (b) Define symmetric and skew symmetric matrix. Express the following matrix as the sum of symmetric and skew symmetric matrices. (10)

$$A = \begin{bmatrix} 1 & 4 & 5 \\ 8 & 5 & 3 \\ 0 & 6 & -2 \end{bmatrix}$$

- (c) Find a  $3 \times 3$  nonsingular matrix A satisfying  $3A = A^2 + AB$ . Hence, find inverse of A using elementary row transformation, where, (13)

$$A = \begin{bmatrix} 2 & 0 & -1 \\ 0 & 0 & -1 \\ -1 & 0 & 1 \end{bmatrix}$$

- Q 6. (a) Reduce the matrix A to its echelon form, then to its canonical form, then to its normal form and hence find its rank, where, (18)

$$A = \begin{bmatrix} 2 & 1 & 5 & 1 & 5 \\ 1 & 1 & -3 & -4 & -1 \\ 3 & 6 & -2 & 1 & 8 \\ 2 & 2 & 2 & -3 & 2 \end{bmatrix}$$

- (b) For which values of 'a' will the following system have (i) no solutions? (ii) exactly one solution? (iii) infinitely many solutions? (17)

$$x + 2y - 3z = 4, \quad 3x - y + 5z = 2, \quad 4x + y + (a^2 - 14)z = a + 2$$

- Q 7. (a) When is a system of non-homogeneous linear equations said to be consistent? Determine whether the following system is consistent or not. (12)

$$x + y + z = 2, \quad x + 2y + 3z = 5, \quad x + 3y + 6z = 11, \quad x + 4y + 10z = 21$$

- (b) Let,  $P_2$  be the vector space of all polynomials with real coefficients of degree 2 or less and  $S = \{P_1, P_2, P_3, P_4\}$  where,  $P_1(x) = -1 + x + 2x^2$ ,  $P_2(x) = x + 3x^2$ ,  $P_3(x) = 1 + 2x + 8x^2$ ,  $P_4(x) = 1 + x + x^2$ . Find a basis of  $P_2$  among the vectors of S. Hence, express the vector that is not in the basis as a linear combination of the basis vectors. (13)

- (c) Define basis of a vector space. Let,  $W = L(S)$  where,  $S = \{(1, -2, 5, -3), (2, 3, 1, -4), (3, 8, -3, 5)\}$  is a subset of  $\mathbb{R}^4$ . Find the basis and dimension of W. (10)

- Q 8. (a) Prove that,  $\text{rank}(A) + \text{nullity}(A) = 4$  (11)

$$A = \begin{bmatrix} 1 & 4 & 5 & 2 \\ 2 & 1 & 3 & 0 \\ -1 & 3 & 2 & 2 \end{bmatrix}$$

- (b) Find all the eigen values of the following matrix and find the corresponding eigen vectors for the largest eigen value. (12)

$$\begin{bmatrix} 2 & 2 & -7 \\ 2 & 1 & 2 \\ 0 & 1 & -3 \end{bmatrix}$$

- (c) Using Cayley-Hamilton theorem, find inverse of the following matrix. (12)

$$A = \begin{pmatrix} 1 & 2 & 2 \\ 3 & 1 & 0 \\ 1 & 1 & 1 \end{pmatrix}$$

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**ME 2127**  
**Solid Mechanics**

Time: 3 hours

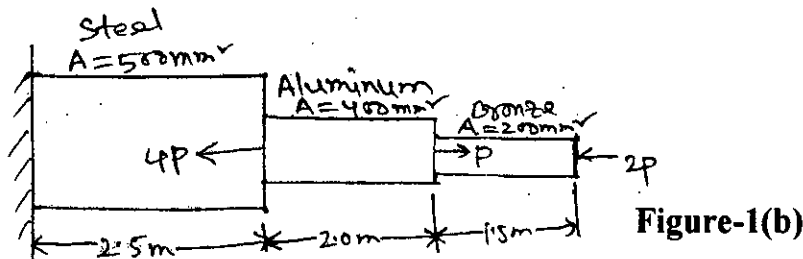
Full Marks: 210

- N.B. (i) Answer **ANY THREE** questions from each section in separate scripts.  
(ii) Figure in the right margin indicates full marks  
(iii) Assume reasonable data if any missing

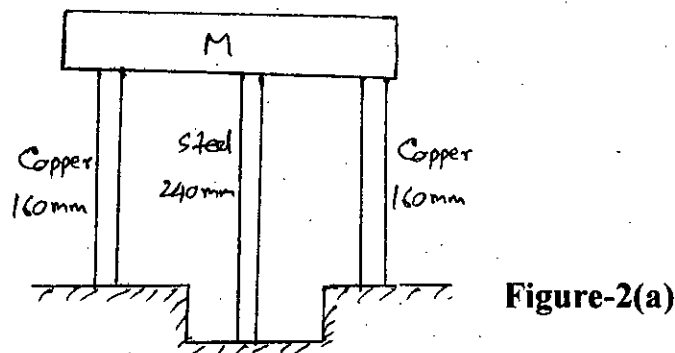
**Section A**

(Answer **ANY THREE** questions from this section in **answer script A**)

- Q 1. (a) Explain the following terms: (08)  
(i) Normal stress (ii) Shearing stress (iii) Hook's law and (iv) Poisson's ratio
- (b) An aluminum rod is rigidly attached between a steel rod and a bronze rod as shown in **Figure-1(b)**. Axial loads are applied at the positions indicated. Find the maximum value of **P** that will not exceed a stress in steel of 140 MPa, in aluminum of 90 MPa or in bronze of 100 MPa. (15)



- (c) A cylindrical steel pressure vessel 400 mm in diameter with a wall thickness of 20 mm, is subjected to an internal pressure of 4.5 MN/m<sup>2</sup>. (i) Calculate the tangential and longitudinal stresses in steel. (ii) To what value may the internal pressure be increased if the stress in the steel is limited to 120 MN/m<sup>2</sup>? (iii) If the internal pressure were increased until the vessel burst, sketch the type of fracture that would occur. (12)
- Q 2. (a) A rigid block of mass **M** is supported by three symmetrically spaced rods as shown, in **Figure-2(a)**. Each copper rod has an area of 900 mm<sup>2</sup>,  $E=120$  GPa and the allowable stress is 70 MPa. The steel rod has an area of 1200 mm<sup>2</sup>,  $E= 200$  GPa and the allowable stress is 140 MPa. Determine the largest mass **M** which can be supported. (17)



- (b) Three rod each of area 250 mm<sup>2</sup>, jointly supported a 7.5 kN load as shown, in **Figure-2(b)**. Assuming that there was no slack or stress in the rods before the load was applied. Find the stress in each rod. Use  $E_{st}= 200$  GPa and  $E_{br}= 83$  GPa. (see figure on next page) (18)

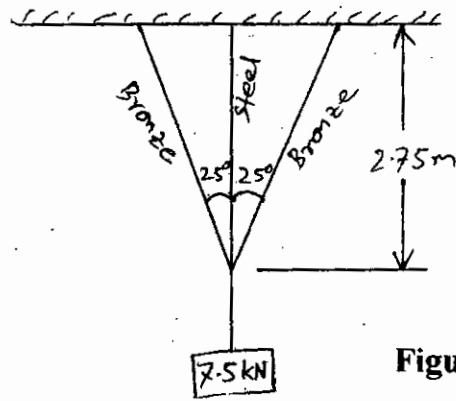


Figure-2(b)

- Q 3. (a) A shaft composed of segments AC, CD and BD is fastened to rigid supports and loaded as shown in Figure-3(a). For bronze,  $G = 35 \text{ GPa}$ ; for Aluminum,  $G = 28 \text{ GPa}$  and for steel,  $G = 83 \text{ GPa}$ . Determine the maximum shearing stress developed in each segment. (18)

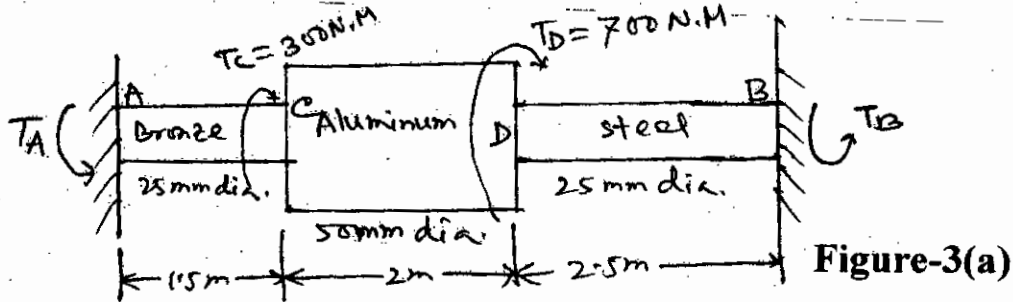


Figure-3(a)

- (b) A rigid bar, hinged at one end, is supported by two identical springs as shown in Figure-3(b). Each spring consists of 20 turns of 10 mm wire having a mean diameter of 150 mm. Compute the maximum shearing stress in the springs. Neglect the mass of the rigid bar. (17)

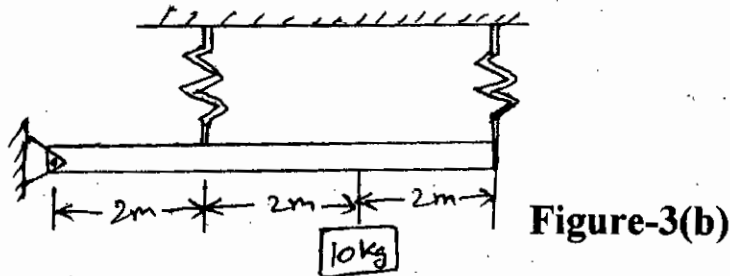


Figure-3(b)

- Q 4. (a) Write the shear and moment equations for the cantilever beam carrying the uniformly varying load and concentrated load as shown in Figure-4(a). Also sketch the shear and moment diagrams. (17)

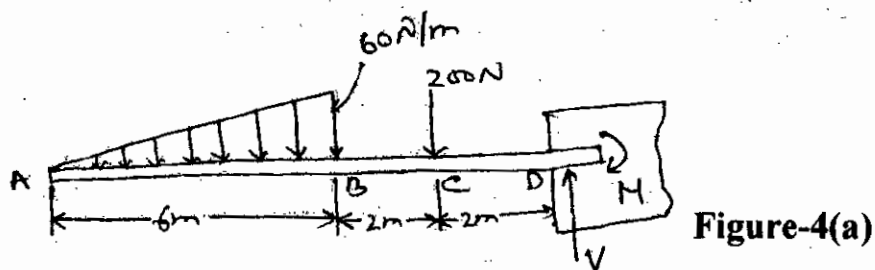


Figure-4(a)

- (b) Without writing shear and moment equations, draw shear and moment diagram for the beam specified as shown in the Figure-4(b). Give the numerical values at all change of loading position and at all points of zero shear. (18)

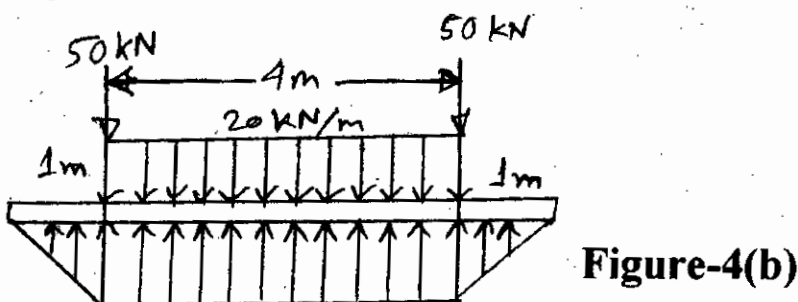


Figure-4(b)

## Section B

(Answer ANY THREE questions from this section in answer script B)

- Q 5. (a) Derive flexure formula with necessary assumption. (10)
- (b) Show that the shearing stress in rectangular section of a beam is distributed parabolically across the depth of the section. (09)
- (c) Find the maximum tensile and compressive flexure stress for the cantilever beam as shown in Figure-5(c). (16)

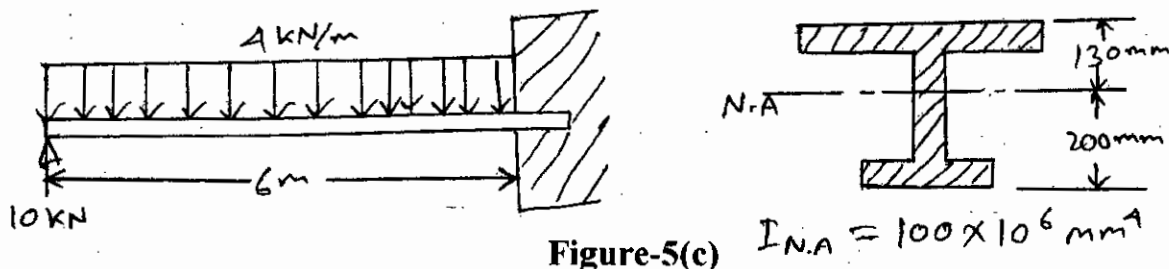


Figure-5(c)

- Q 6. (a) The frame as shown in Figure-6(a) is supported by a hinge at A and a roller at E. Compute the horizontal and vertical components of the hinge forces at B and C as they act upon member AC. (18)

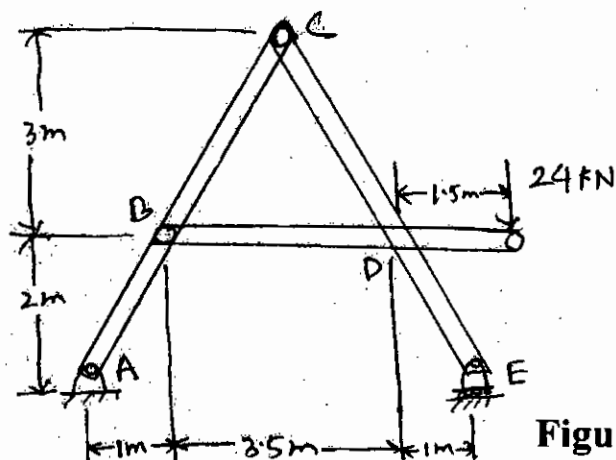


Figure-6(a)

- (b) A cylinder weighs 400 kg and is lifted by a pair of tongs as shown in Figure-6(b). Determine the forces exerted at D and C on the tong BCD. (17)

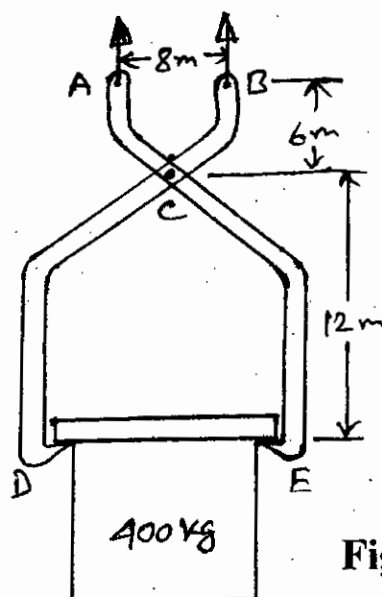
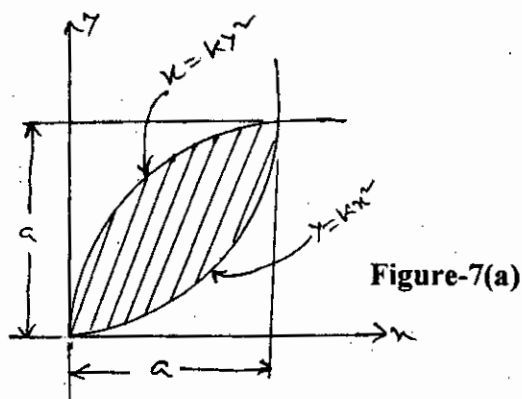


Figure-6(b)

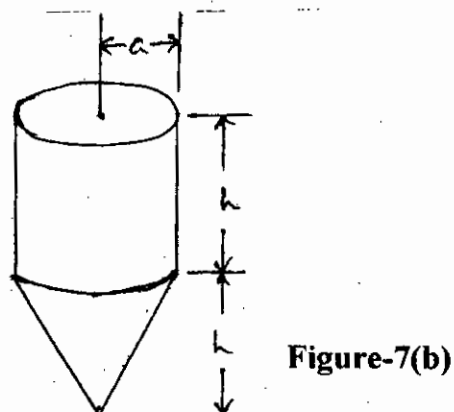
Q 7. (a) Determine by direct integration the centroid of the area as shown in **Figure-7(a)**.

(18)



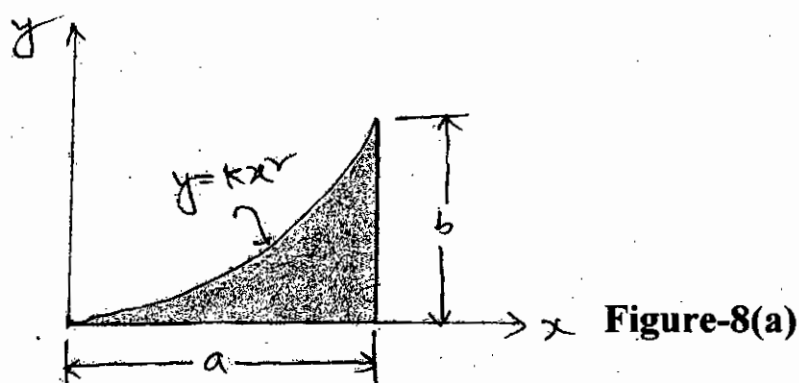
(b) A cone and a cylinder of the same radius 'a' and height 'h' are attached as shown in **Figure-7(b)**. Determine the location of the centroid of the composite body.

(17)



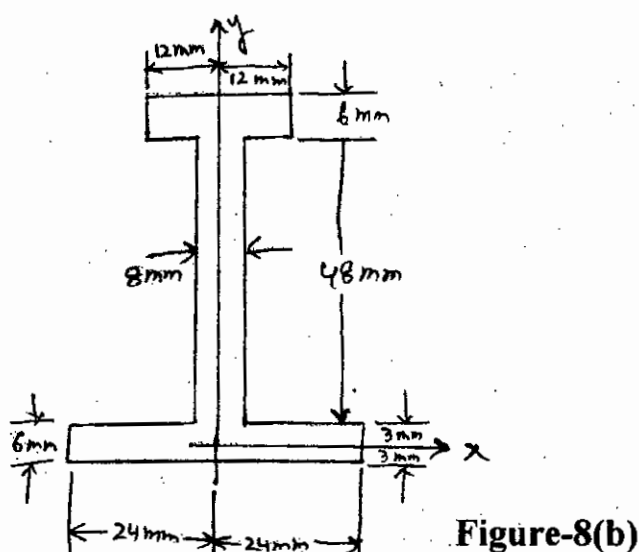
Q 8. (a) Determine the moment of inertia of the shaded area shown in **Figure-8(a)** with respect to each of the coordinate axes. Also determine the radius of gyration of the shaded area with respect to each of the coordinate axes.

(20)



(b) Determine the moment of inertia of the area as shown in **Figure-8(b)** with respect to x and y-axis.

(15)





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**Hum 2127**  
**Accounting and Economics**

Time: 3 hours

Full Marks: 210

- N.B. (i) Answer **ANY THREE** questions from each section in separate scripts.  
(ii) Figure in the right margin indicates full marks  
(iii) Assume reasonable data if any missing

**Section A**

(Answer **ANY THREE** questions from this section in **answer script A**)

- Q 1. (a) Define Accounting equation. State the elements of accounting equation. (15)  
(b) Who are the users of Accounting Information? Explain. (10)  
(c) Describe the rules of "Debit and Credit" of accounts. (10)

- Q 2. (a) What is meant by accounting cycle? (05)  
(b) Describe the steps of an accounting cycle. (12)  
(c) Some transactions for Rehan, in his first month of business are as follows: (18)

January 1– Invested tk. 50,000 cash and furniture of 75,000 in business  
January 4– Purchased used car for tk. 90,000 cash for use in business  
January 8– Hires an administrative assistant  
January 10– Purchased supplies on account for tk. 15,000  
January 13– Billed customer tk. 21,000 for service provided  
January 17– Paid tk. 3,000 cash for advertising and 5,000 as insurance premium  
January 21– Received tk. 3,000 cash from customer billed on January 13  
January 25– Payments to Account Payable tk. 1,000  
January 29– Withdraw tk. 2,000 cash for personal use of owner

Required: Journalize above transactions.

- Q 3. (a) What is cost? (05)  
(b) Classify cost in relation to accounting period. (10)  
(c) Define trial balance. What are the errors that cannot be detected by a trail balance? (20)

- Q 4. The unadjusted trail balance of Nupur Company as of December 31, 2009 follows: (35)

| Name of Accounts                   | Debit (tk.)     | Credit (tk.)    |
|------------------------------------|-----------------|-----------------|
| Cash                               | 11,000          |                 |
| Supplies                           | 4,500           |                 |
| Equipment                          | 1,50,000        |                 |
| Accumulated depreciation–equipment |                 | 15,000          |
| Unearned membership fees           |                 | 24,000          |
| Notes Payable                      |                 | 50,000          |
| Capital                            |                 | 58,250          |
| Withdrawals                        | 30,000          |                 |
| Fees received                      |                 | 90,000          |
| Salary expenses                    | 38,000          |                 |
| Interest expenses                  | 3,750           |                 |
| <b>Total</b>                       | <b>2,37,250</b> | <b>2,37,250</b> |

(continuing ... ..)

Additional data:

- (a) On 31 December, employees have earned tk. 800 but unpaid
- (b) Supplies on hand are tk. 1,800
- (c) Accrued interest is tk. 1,250
- (d) From the unearned membership fees tk. 16,000 remains unearned at December 31
- (e) In addition to the membership fees included in the revenue account balance, the company has earned another tk. 12,000 fees that will be collected on January 21
- (f) Depreciation expense for the year is tk. 15,000.

Required: (1) Prepare income statement for the year ended December 31.  
(2) Prepare balance sheet as on that date.

### **Section B**

(Answer **ANY THREE** questions from this section in **answer script B**).

- Q 5. (a) How is Economics a science? Explain the idea of Economics, microeconomics and macroeconomics with appropriate example. (15)
- (b) "People Face trade-off"—use a Production Possibilities Frontier (PPF) to illustrate society's trade-off between two "goods"—a clean environment and the quantity of industrial output. Show what happens to the Frontier if engineers develop a new way of producing electricity that emits fewer pollutants. (20)
- Q 6. (a) What is demand? What determines the quantity of a goods that buyers demand? (15)
- (b) On a supply-and-demand diagram, show equilibrium price, equilibrium quantity. If the demand is inelastic, how will an increase in price change total revenue. Explain. (20)
- Q 7. (a) What is Gross Domestic Product (GDP), Real GDP and GDP growth? (15)
- (b) Explain any one method of calculating GDP. (10)
- (c) What is private savings, public savings and national savings? Explain. (10)
- Q 8. (a) What is Production? Discuss the different factors of production. (15)
- (b) Draw the marginal-cost and average-total-cost curves for a typical firm. Explain why the curves have the shapes that they do and why they cross where they do? (20)

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**MSE 2103**  
**Thermodynamics of Materials**

Time: 3 hours

Full Marks: 210

- N.B. (i) Answer **ANY THREE** questions from each section in separate scripts.  
(ii) Figure in the right margin indicates full marks

**Section A**

(Answer **ANY THREE** questions from this section in **answer script A**)

- Q 1. (a) State and explain the first law of thermodynamics and show that, for an infinitesimal change in the condition of the system,  $dU = \delta Q + \delta W + \delta W'$ , where the symbols have their usual meanings. (15)
- (b) Prove that, heat absorbed in any reversible isothermal process is the maximum. (10)
- (c) Write total differential equation of the function  $z = 17x^4y + 22xy^5$  and then, using Maxwell relation, prove that,  $z$  is state function. (10)
- Q 2. (a) What are the extensive and intensive properties of thermodynamic systems? Explain with example. (10)
- (b) Classify the following thermodynamic systems: (5×2)
- i. A solid bar of copper
  - ii. A glass of ice water
  - iii. A yttria-stabilized zirconia furnace tube
  - iv. A Styrofoam coffee cup
  - v. A eutectic alloy water turbine blade rotating at 1000 rpm
- (c) What is entropy? Show that, the entropy of a system may increase or decrease for a given process, but the entropy of the universe can only increase. (10)
- (d) Discuss the limitation of thermodynamics. (05)
- Q 3. (a) Show that, heat is not a state variable. (15)
- (b) Express the change in enthalpy as a function of entropy and volume. (10)
- (c) Compute the change in Gibbs free energy of magnesia (MgO) when 1 mol is heated from 298 to 1300 K at 1 atm pressure. (10)
- Given,  $C_P(\text{MgO}) = 48.99 + 3.43 \times 10^{-3} T - 11.43 \times 10^{-5} T^2 \text{ J/mol-K}$ ;  $S_{298}^\circ = 26.9 \text{ J/mol-K}$ .
- (Use Table 1 relationships, if required)
- Q 4. (a) Differentiate between steady state and equilibrium state with an example. (10)
- (b) Compute the change in entropy when 1 mole of an ideal gas expands freely to twice its volume. (10)
- (c) State the alternate condition for spontaneous change and criteria for equilibrium. (10)
- (d) State the third law of thermodynamics using a suitable example. (05)

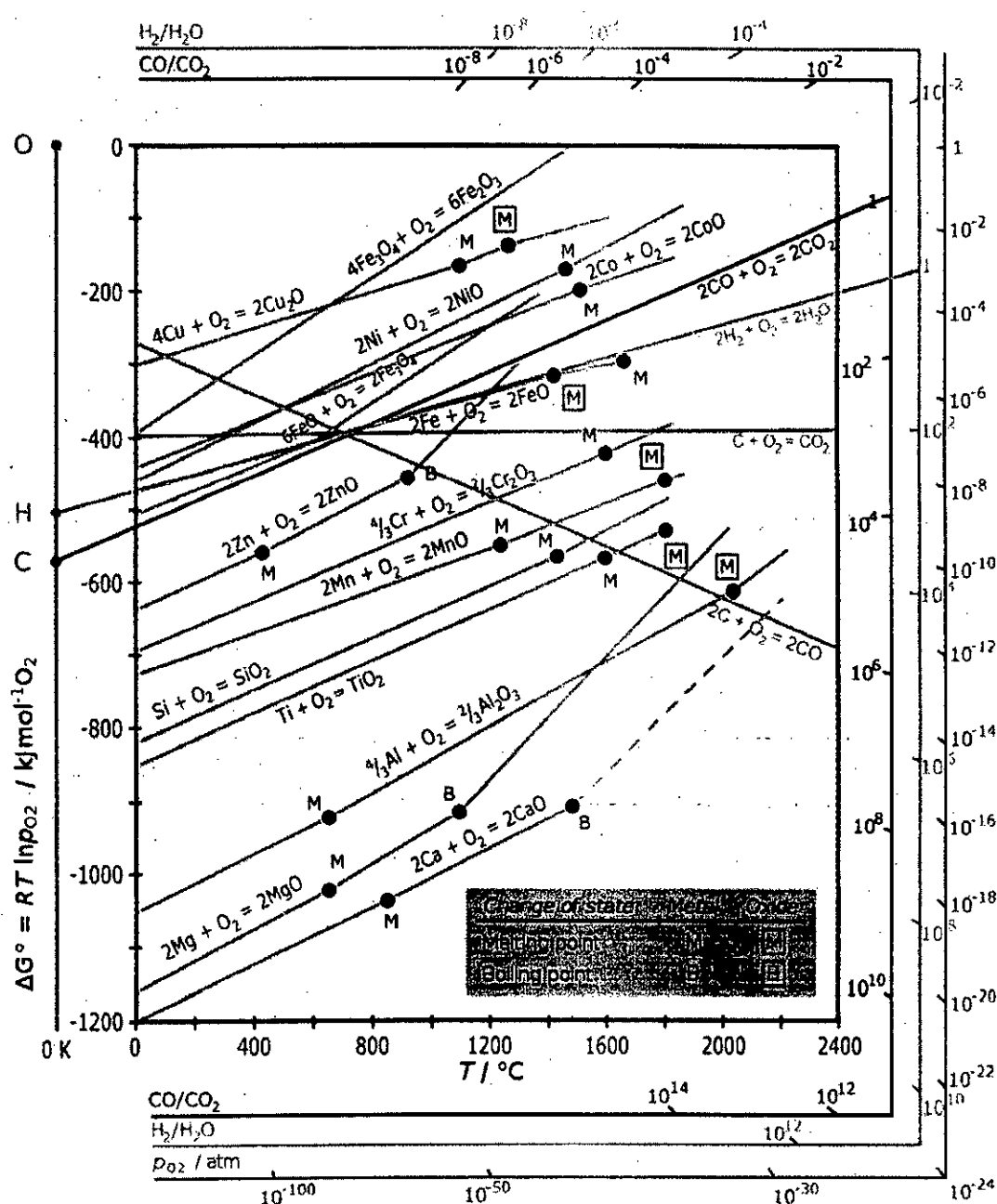
### Section B

(Answer **ANY THREE** questions from this section in **answer script B**)

- Q 5. (a) Derive the Clausius-Clapeyron equation for a unary system in which sublimation occurs and deviation from ideal gas behavior is negligible. (15)
- (b) Explain the calculation of chemical potential surfaces for the construction of unary phase diagrams. (10)
- (c) The normal boiling point of zinc is 1180 K. If the normal heat of vaporization at the normal boiling point is 104 kcal/mol and the heat capacities at constant pressure of liquid and gaseous zinc are 31.38 and 20.79 cal/deg-mol, respectively, calculate the vapor pressure at 1073 K. (10)
- Q 6. (a) Describe the methods of the evaluation of partial molar properties in a binary solution from the total molar properties. (15)
- (b) Deduce the form of Gibbs-Duhem equation which can be applied to the mixing process. Explain how this equation may be used to calculate partial molar properties of one component in a binary solution knowing those of the other component. (20)
- Q 7. (a) Point out the characteristics of the plots representing mixing behavior of an ideal solution with neat sketches. (13)
- (b) Suppose, a binary solution consists of components A and B. Derive a relation from which activity of component A can be measured if activity of component B is known. (10)
- (c) Compare the phase diagram of a pure substance in (T, P) space to its phase diagram in (V, P) space. (12)
- Q 8. (a) Define variance of a reacting system. Deduce the relation for van't Hoff isotherm using a general chemical reaction:  $lL + mM = rR + sS$ . How does this relation relate to the direction of the reaction? (13)
- (b) Consult the Ellingham diagram in Figure 8(b) to answer the following questions: (3×4)
- Aluminium can be used to reduce titanium oxide at 1000°C whereas chromium cannot be used. Why?
  - Explain why the CO<sub>2</sub> line is horizontal on the diagram but the CO line has a negative slope.
  - Why is carbon an energetic reducing agent at high temperatures?
- (c) Deduce a relation showing the dependence of equilibrium constant for any chemical reaction on temperature. (10)

**Table 1: Thermodynamics State Functions Expressed in Terms of the Independent variables Temperature and Pressure**

|               |                                                     |
|---------------|-----------------------------------------------------|
| $V = V(T, P)$ | $dV = V\alpha dT - V\beta dP$                       |
| $S = S(T, P)$ | $dS = \frac{C_P}{T} dT - V\alpha dP$                |
| $U = U(T, P)$ | $dU = (C_P - PV\alpha) dT + V(p\beta - T\alpha) dP$ |
| $H = H(T, P)$ | $dH = C_P dT + V(1 - T\alpha) dP$                   |
| $F = F(T, P)$ | $dF = -(S + PV\alpha) dT + PV\beta dP$              |
| $G = G(T, P)$ | $dG = -S dT + V dP$                                 |



**Figure 8(b): The Ellingham Diagram**